



MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING & INFORMATION TECHNOLOGY

**AGRILINK: A DIGITAL MARKETPLACE PLATFORM FOR DIRECT
FARMER-TO-BUYER AGRICULTURAL TRADE, SUPPLIER
CONNECTIVITY, CROP ANALYTICS, AND SUBSIDY ACCESS**

BY

NAME: MIGIRO ASA JOSHUA

REG. No: CIT-223-057/2022

SUPERVISOR: NICK GACHUI

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Project Proposal submitted in partial fulfillment of the requirements of Bachelor of
Science in Computer/Information Technology

DECLARATION

I hereby declare that this Project Proposal is my own work and has, to the best of my knowledge, not been submitted to any other institution of higher learning.

Student : _____

Registration Number:

Signature:

Date:.....

This Project Proposal has been submitted as a partial fulfillment of requirements for the Bachelor of Science in Computer/Information Technology of Multimedia University of Kenya with my approval as the University supervisor.

Supervisor: _____

Signature:
.....

Date:

DEDICATION

This project is dedicated to the hardworking smallholder farmers across Kenya and beyond, who feed nations yet often receive the least fair compensation for their labor. May technology serve as a bridge to equity and empowerment.

I also dedicate this project to my family and friends who supported me throughout my studies in MMU and enabled me to work on this project smoothly.

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ABSTRACT

The agricultural sector in Kenya is characterized by fragmented market access and limited bargaining power for smallholder farmers. Many farmers rely on brokers and middlemen who exploit information asymmetry, resulting in low farm-gate prices and reduced farmer incomes. In addition, access to verified farm input suppliers remains a challenge, exposing farmers to counterfeit or substandard seeds, fertilizers, and agrochemicals that negatively affect productivity.

This project proposes **AgriLink**, a web-based platform designed to (1) connect farmers directly to buyers to support fair pricing and transparent trade, and (2) connect farmers to verified agricultural input suppliers through a supplier directory and request system. The system is developed using the **Agile methodology** over a seven-week timeline and implemented using a beginner-friendly full-stack web technology stack. Primary and secondary data collection methods are used to identify user requirements, while system validation is performed through unit testing, integration testing, and user acceptance testing.

The developed solution demonstrates that a lightweight digital platform can reduce dependency on brokers, improve market visibility for farmers, and enhance access to quality inputs. Future work recommends extending AgriLink with **crop analytics dashboards** and **transport/logistics coordination** to improve decision-making and last-mile delivery.

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CHAPTER 1: INTRODUCTION

1.1 Background of Study

Agriculture is one of the most important sectors in Kenya and plays a central role in economic development, employment, and national food security. It contributes significantly to the country's Gross Domestic Product (GDP) and supports livelihoods for a large portion of the rural population. Smallholder farmers form the majority of agricultural producers and are responsible for much of the food consumed locally. Despite their importance, many smallholder farmers continue to face structural barriers that reduce their earnings and limit their ability to grow their farming enterprises.

One major challenge in the agricultural value chain is **market access**. Many farmers are unable to reach buyers directly due to a lack of market information, distance to markets, and weak bargaining power. As a result, farmers often depend on brokers or intermediaries who purchase produce at low farm-gate prices and resell at higher prices in urban markets. This arrangement creates price exploitation and reduces farmers' profitability.

Another challenge is **access to quality agricultural inputs**. Farmers require seeds, fertilizers, pesticides, and farm tools to maximize productivity. However, input supply markets are often dominated by informal vendors, and farmers may purchase counterfeit or low-quality products that lead to poor yields and financial loss. Additionally, suppliers may not reach remote farming communities, limiting access and increasing costs.

With increased adoption of mobile phones and internet connectivity in Kenya, digital platforms present an opportunity to improve market linkages and reduce the inefficiencies in traditional trading systems. Several AgriTech solutions have attempted to address market access and input supply, but there remains a gap for a platform that integrates direct farmer-to-buyer sales with verified supplier connections in a simple and accessible system. Therefore, this project proposes **AgriLink**, a web-based platform aimed at strengthening farmer income opportunities through direct market connections and improved access to input suppliers.

1.2 Problem Statement

Smallholder farmers in Kenya frequently sell their produce through brokers and intermediaries due to limited access to direct markets. Brokers often determine prices unilaterally, leading to unfair compensation for farmers. Farmers generally lack visibility into prevailing market prices and do not have structured channels for reaching reliable buyers such as institutions, retailers, restaurants, and individual consumers.

At the same time, farmers face difficulties accessing verified input suppliers. Many farmers purchase seeds, fertilizers, and pesticides from informal vendors without assurance of quality. This increases the risk of counterfeit inputs, reduced yields, and

wasted financial resources. There is no widely accessible platform that simultaneously enables farmers to list produce for direct sale and also discover and communicate with verified suppliers.

Therefore, there is a need for a digital platform that enables transparent farmer-to-buyer transactions and connects farmers with agricultural input suppliers in order to improve farmer income and productivity.

1.3 Aim of the Study

The aim of this project is to design, develop, and implement **AgriLink**, a web-based platform that enables **direct farmer-to-buyer trade** and connects farmers to **verified agricultural input suppliers**.

1.3.1 Research Objectives

The specific objectives of the study are:

1. To analyze challenges faced by smallholder farmers in accessing direct markets and quality input suppliers.
2. To design a web-based system that supports direct produce listings, buyer discovery, and supplier connection features.
3. To implement core platform modules including authentication, farmer listings, buyer ordering, supplier catalogs, and request workflows.
4. To test and evaluate the system using unit testing, integration testing, and user acceptance testing.
5. To provide recommendations for future improvements including crop analytics and transport/logistics integration.

1.4 Significance/Justification of the Study

To Farmers: AgriLink provides farmers with a direct channel to buyers, improving market access and allowing farmers to receive fairer prices. It also reduces exposure to counterfeit farm inputs by linking farmers to verified suppliers.

To Buyers: Buyers gain access to farm produce directly from farmers, improving freshness, traceability, and potentially lowering prices due to reduced intermediary costs.

To Suppliers: Input suppliers benefit from improved visibility and a structured platform to reach farmers, enabling growth in customer base and improved communication.

To Government and Policy Makers: Although not directly integrating subsidies in scope, AgriLink supports national goals of agricultural modernization by improving market efficiency and farmer empowerment.

To Academia and Research: The project contributes a practical AgriTech case study and demonstrates the application of software engineering principles to solve real-world problems in agriculture.

1.5 Scope

In Scope

- Web-based system accessible via modern browsers (desktop and mobile)
- User registration and login with role-based access: Farmer, Buyer, Supplier, Admin
- Farmer profile management
- Farmer product listing (CRUD) with images
- Buyer product browsing with search and filters
- Buyer order placement and order history
- Farmer order management (accept/reject/fulfill)
- Supplier profiles and input product catalog (CRUD)
- Farmer requests/inquiries to suppliers and supplier responses
- Admin moderation and basic management functions
- Deployment for localhost development and live deployment for demonstration

Out of Scope (Current Version)

- Crop analytics dashboard (future work)
- Transport/logistics module (future work)
- Integrated payments (future work; orders can be coordinated externally e.g., M-Pesa)
- Native mobile applications (future work)
- Automated subsidy application and government database integration

1.6 Assumptions

1. Users have access to a smartphone or computer and internet connection.
2. Farmers and buyers can understand basic interactions in a web application.
3. Suppliers will provide accurate business and product information.

4. The system will be used within the Kenyan agricultural context.
5. The seven-week development timeline is sufficient to build a functional MVP for the two scoped modules.

1.7 Limitations

Limitation/Challenge	Impact	Countermeasure
Solo development and limited time	Reduced feature breadth	Prioritize MVP features and apply Agile sprints
Beginner skill level	Slower implementation	Use beginner-friendly stack and component libraries
No integrated payment processing	Less complete e-commerce flow	Use order confirmation + external payment instructions
Limited access to real farmers for UAT	Testing realism may be reduced	Use simulated users + a small sample of real respondents where possible